CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 7: Ratio and Proportion

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the duplicate ratio and the sub-duplicate ratio of $9 : 16$.

2. Find the compound ratio of $3 : 4$, $8 : 9$ and $3 : 2$.

3. If $x : y = 4 : 7$, find $(3x + 2y) : (5x + y)$.

4. Find the fourth proportional to 4, 7 and 12.

5. Find the mean proportional between 8 and 18.

6. Find the third proportional to 9 and 12.

7. What number must be subtracted from each of 7, 17 and 47 so that the results are in continued proportion?

8. If $\dfrac{5x + 2y}{5x - 2y} = \dfrac{4}{3}$, find $x : y$.

9. Solve $\dfrac{\sqrt{x+7} + \sqrt{x-8}}{\sqrt{x+7} - \sqrt{x-8}} = \dfrac{5}{3}$.

10. If $\dfrac{a}{b} = \dfrac{c}{d}$, prove that $\dfrac{a+c}{b+d} = \dfrac{a}{b}$.

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Answer Key

1. $81 : 256$ and $3 : 4$
2. $72 : 72 = 1 : 1$
3. $26 : 27$
4. 21
5. 12
6. 16
7. $(17 - x)^2 = (7 - x)(47 - x)$ gives $x = 2$
8. $\frac{5x}{2y} = 7$, so $x : y = 14 : 5$
9. $\frac{\sqrt{x+7}}{\sqrt{x-8}} = 4$, so $x + 7 = 16(x - 8)$ and $x = 9$
10. With $a = bk$, $c = dk$: LHS $= \frac{k(b+d)}{b+d} = k =$ RHS

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